

Mohamed Zeina

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EXPERIENCE

Embedded Systems & Full Stack Software Engineer (Contract)

August 2025 – Apr. 2026

Energco, ZAIM Teknopark

Istanbul, TR

- Sole developer of the entire full-stack system: architected, built, and deployed the complete backend and frontend for a multi-drone platform coordinating 5 UAVs with real-time 30 FPS video streaming, live telemetry via WebSockets, and fault-tolerant reconnection and failsafe handling
- Developed a Ground Control Station using React and Python FastAPI with real-time map interface, manual flight controls, and remote GPS navigation
- Implemented YOLOv8 object detection with PID-controlled flight for real-time subject tracking; calculated target GPS from gimbal angles, GPS, and rangefinder data to autonomously dispatch mini drones
- Configured bridged wireless network between GCS and drones; set up embedded Linux on Raspberry Pi for video streaming and MAVLink communication

Teaching Assistant for Introduction to Computing

Sep. 2021 – Jan. 2024

Sabanci University

Istanbul, TR

- Tutored over 50 students weekly in C++ programming; designed and evaluated 30+ assignments and exams aligned with course learning objectives

Full Stack Software Engineer Intern

July 2019 – Sep. 2019

Interact Technologies

Istanbul, TR

- Implemented TCP/IP protocol achieving 99.9% data transmission success and reduced GUI load time by 40% through code optimization
- Designed a NoSQL document database schema supporting 500+ patient records, enabling downstream reporting

EDUCATION

Sabanci University

Istanbul, TR

Master of Science in Computer Science and Engineering (M.Sc.CSE)

Sep. 2021 – Jan. 2024

- GPA: 3.91/4.0 | Full Scholarship | Thesis: Built a framework combining GANs, SMOTE, and differential privacy to preserve household privacy in smart meter data

Sabanci University

Istanbul, TR

Bachelor of Computer Science and Engineering (B.Sc.CSE)

Sep. 2016 – June 2020

- GPA: 3.73/4.0 | Half Tuition Scholarship | Degree of High Honor (2016–2017, 2018–2020)

PROJECTS

[Noblecart](#) | [GitHub](#) | *Node.js, Express.js, MongoDB, EJS, Three.js, Stripe, Cloudinary*

- Built a full-stack MVC e-commerce app with session auth (bcrypt, CSRF, Helmet hardening), role-scoped admin controls, and Cloudinary storage for images and GLB 3D models
- Implemented an interactive Three.js 3D product viewer, a slide-out cart drawer with live quantity controls via Fetch API, Stripe Checkout payments, and server-side PDF invoice generation with PDFKit

[Hearsay](#) | [GitHub](#) | *Next.js, TypeScript, PostgreSQL, NextAuth, Tailwind, Prisma, Zod*

- Built a full-stack threaded discussion app with GitHub OAuth, recursively nested comments with reply-aware delete logic, live typeahead search, and authorization-checked, Zod-validated mutations
- Architected on the Next.js App Router with React Server Components and Suspense streaming for parallel post/comment/sidebar loading, React `cache()` for per-render query deduplication, and a custom Tailwind design system

[Dispatches](#) | [GitHub](#) | *Node.js, Express.js, MongoDB, GraphQL, JWT, Cloudinary, React*

- Built a GraphQL API (Express, Mongoose) with per-resolver JWT authorization (1h expiry, bcrypt hashing), server-side validation, paginated feeds, ownership-enforced post CRUD, idempotent likes via `$addToSet/$pull`, and Multer + Cloudinary image uploads
- Modeled recursively nested comment threads as an indexed adjacency list with BFS cascade-delete; built a React optimistic-UI layer (likes and comments reconcile against the server with rollback and swap-stable keys) and a modal-as-route post view that syncs counts across feed and profile without refetching

[JSNote](#) | [GitHub](#) | *React, TypeScript, Vite, esbuild-wasm, Monaco, Redux, npm workspaces*

- Published `jsnote-zeina` to npm as a CLI-distributed in-browser JS/JSX notebook with Monaco IntelliSense via `@typescript/ata`, esbuild-wasm bundling with unpkg resolution, sandboxed iframe execution, and cumulative cell scope sharing
- Architected as a three-package npm workspace with Redux/Immer state, debounced persistence, and runtime-error cell attribution via source-map-js, covered by Vitest and Playwright

SKILLS

Languages: C++, C#, Python, Java, JavaScript, TypeScript, SQL

Frontend: React, Redux, Next.js, Tailwind CSS, Three.js, Vite

Backend & APIs: Node.js, Express, FastAPI, GraphQL, RESTful APIs, WebSockets, NATS, Microservices, JWT, Stripe

Databases & ORMs: MongoDB, PostgreSQL, Redis, Mongoose, Prisma

DevOps & Cloud: Docker, Nginx, Kubernetes, AWS (EC2, RDS, S3), Google Cloud (GKE), CI/CD (GitHub Actions, Travis CI)

Testing: Vitest, Playwright, Jest

AI & LLMs: Claude Code, Multi-agent Orchestration, Prompt Engineering, LLM Integration

Embedded & Computer Vision: Raspberry Pi, MAVLink, UDP Streaming, OpenCV, YOLOv8

Spoken: Arabic (Native), English (Fluent)